

Firewall configuration Theory

1. Install ufw on your Linux system (if not already installed) and enable it, then verify its status using the appropriate commands.

ANS :

❖ **Install UFW**

- sudo apt update
- sudo apt install ufw -y

❖ **Check UFW Status**

- sudo ufw status

❖ **Enable UFW**

- sudo ufw enable

❖ **Verify:**

- sudo ufw status verbose

2. Configure ufw to allow incoming SSH connections only on port 22 and block all other incoming connections.

ANS :

❖ **Set default incoming policy to deny:**

- sudo ufw default deny incoming

❖ **Allow SSH:**

- sudo ufw allow 22/tcp

❖ **Verify rules:**

- sudo ufw status numbered

❖ **Expected result:**

- 22/tcp ALLOW IN Anywhere

3. Add a rule in ufw to allow outgoing HTTP and HTTPS traffic (ports 80 and 443), but block all other outbound connections.

Hint: Use 'ufw allow out' and 'ufw deny out' commands to control outbound traffic.

ANS :

❖ **Set default outgoing policy:**

- sudo ufw default deny outgoing

❖ **Allow HTTP:**

- sudo ufw allow out 80/tcp

❖ **Allow HTTPS:**

- sudo ufw allow out 443/tcp

- ❖ Verify:
 - `sudo ufw status verbose`
 - ❖ Expected result:
 - 80/tcp ALLOW OUT Anywhere
 - 443/tcp ALLOW OUT Anywhere
4. List all currently active ufw rules on your system and save the output to a file named `firewall_rules.txt`.

ANS :

- ❖ Display all rules:
 - `sudo ufw status numbered`
 - ❖ Save output:
 - `sudo ufw status numbered > firewall_rules.txt`
 - ❖ Verify file:
 - `cat firewall_rules.txt`
5. Write a short explanation (3-4 lines) describing how your current ufw configuration would impact running a local web server (like a development server for a Zomato-style food delivery app).

ANS :

- Current UFW configuration allows incoming SSH connections only on port 22 and blocks all other incoming traffic. As a result, a local web server running on port 80 or 443 will not be accessible from other systems unless additional firewall rules are added. Outbound HTTP and HTTPS traffic is allowed, so the server can download updates and communicate with external web services. This configuration improves security but restricts public access to the web application.